Linu Anil Teresa

443-800-6928 | lteresa@umd.edu | linkedin.com/in/linuteresa | github.com/linuteresa

Summary

M.S. Data Science candidate (GPA 4.0/4.0) and software engineer who takes ML, data, and agentic-AI systems from research to production. 2+ years shipping scalable backend services (Java/Spring Boot) plus hands-on work with LLMs, multi-agent orchestration, and big-data pipelines. First-author IEEE publication. Open to **AI/ML Engineering, Software Engineering, and Data Science** roles.

Education

University of Maryland — College Park, MD
Master of Science in Data Science, GPA: 4.0/4.0 — *Aug. 2025 – May 2027*

Saintgits College of Engineering — Kerala, India
Bachelor of Science in Computer Science and Engineering — *Jul. 2019 – Jun. 2023*

Experience

Machine Learning Research Intern — May 2026 – Present
Vitreous State Laboratory, The Catholic University of America — *Washington, DC*

- Owned end-to-end Python ML pipelines to develop, test, and validate models on real scientific datasets; built a reproducible evaluation stack (pandas, scikit-learn) with cross-validation and significance testing

Software Engineer (Packaged App Development Associate) — Jul. 2023 – Jul. 2025
Accenture — *Kochi, India*

- Shipped scalable Java (Spring Boot) microservices and REST APIs for enterprise analytics dashboards used by 1,000+ daily users, improving reliability under production load
- Accelerated backend response 35–40% via SQL query optimization and caching, and lifted test coverage to 85% (JUnit, Karate) – cutting production defects 60%
- Owned features across the full lifecycle – design, code review, and CI/CD deployment – on an Agile team

Research Intern (IoT & Systems) — Dec. 2021 – Dec. 2022
International Institute of Information Technology (IIIT Hyderabad) — *Hyderabad, India*

- Delivered a real-time data pipeline and dashboard for distributed sensor data that secured Government of Kerala funding to scale the project

Projects

Bayes Execution Engine | *Python, LangGraph, MCP, pgmpy, LLMs*

- Built a deterministic Plan-and-Execute multi-agent orchestration engine (LangGraph, MCP) with Bayesian probabilistic modeling (pgmpy) to eliminate hallucination loops and resolve data conflicts across local LLMs

Big Data Analytics | *Apache Spark, Airflow, Dask, MongoDB, Neo4j, Redis*

- Built 7 big-data projects spanning batch/stream processing (Spark, Dask), workflow orchestration (Airflow), and storage across relational, document (MongoDB), and graph (Neo4j) databases

Data Pipeline & Analytics Framework: Formula-One Historical Data | *Python, pandas*

- Built an analytics pipeline over Formula-One historical race data – ingestion, feature engineering, and visualization – to quantify overtaking difficulty and surface performance insights

Technical Skills

Data Infrastructure: SQL, Snowflake, AWS, Azure, Apache Spark, Airflow
Languages: Python, Java, SQL, JavaScript, TypeScript
AI/ML & Agents: LLMs/GenAI, LangGraph, LangChain, MCP, Multi-Agent Orchestration, TensorFlow, scikit-learn
Data & Databases: pandas, NumPy, Dask, ETL/Pipelines, MongoDB, Neo4j, Redis, SQL Server

Publications

L. A. Teresa *et al.*, "Enhancing Children's Learning Experience: Interactive and Personalized Video Learning with AI Technology," **IEEE RASSE 2023 (First author)**.